Program : **Diploma in Tool & Die Engineering**	
Course Code : **5112B**	Course Title: **Robotics & Automation**
Semester : **5**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L:4, T:0, P:0)**	Periods per semester: **60**

Course Objectives:

- To introduce the basic concepts, parts of robots and types of robots
- To make the student familiar with the various drive systems for robot, sensors and their applications in robots and programming of robots
- To select the robots according to its usage
- To discuss about the various applications of robots, justification and implementation of robot
- To conceptualize automation and understand applications of robots in various industries

Course Prerequisites:

Topic	Course code	Course name	Semester
Engineering Science		Engineering Mechanics	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the robot anatomy, various robotic actuators and controllers	13	Understanding
CO2	Summarize various types of sensors and concepts on robot vision system	15	Understanding
CO3	Illustrate the concepts of robot programming languages and various methods of robot programming	15	Applying

CO4	Explain automation & the various applications of robots	15	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	3	2					
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain the robot anatomy, various robotic actuators and controllers**		
M1.01	Define robot and its functions	3	Understanding
M1.02	Identify the parts of a robot	3	Understanding
M1.03	Classify robotic joints	3	Understanding
M1.04	Describe various characteristics of a robot	4	Understanding
Contents: **Fundamentals of Robotics**: Introduction; Definition; Robot anatomy (parts) and its working; Robot Components: Manipulator, End effectors; Construction of links, Types of joints; Classification of robots; Cartesian, Cylindrical, Spherical, Scara, Vertical articulated; Structural Characteristics of robots; Mechanical rigidity; Effects of structure on control work envelope and work Volume; Robot work Volumes, comparison; Advantages and disadvantages of robots.			
CO2	**Summarize various types of sensors and concepts on robot vision system.**		
M2.01	Identify different actuators and drives	5	Understanding
M2.02	Explain robotic controls	5	Understanding
M2.03	Familiarize with the most common robot sensors & its applications	5	Understanding
	Series Test I	1	

Contents:

Robotic Drive System and Controller: Actuators; Hydraulic, Pneumatic and Electrical drives; Linear actuator; Rotary drives; AC servo motor; DC servo motors and Stepper motors; Conversion between linear and rotary motion; Feedback devices; Potentiometers; Optical encoders; DC tachometers; Robot controller; Level of Controller; Open loop and Closed loop controller; Microprocessor based control system; Robot path control: Point to point, Continuous path control and Sensor based path control.

Sensors: Requirements of a sensor; Principles and Applications of the following types of sensors: Position sensors (Encoders, Resolvers, Piezo Electric); Range sensors (Triangulation Principle, Structured lighting approach); Proximity sensing; Force and torque sensing.

CO3	**Illustrate the concepts of robot programming languages and various methods of robot programming**		
M3.01	Explain the basics and various components of machine vision systems	3	Understanding
M3.02	Describe the applications of machine vision systems	2	Understanding
M3.03	Identify different programming methods used in robots	3	Understanding
M3.04	Define various commands used in robotic programs	3	Understanding
M3.05	Use simple programmes to simulate robot movements.	4	Applying

Contents:

Introduction to Machine Vision: Robot vision system (scanning and digitizing image data); Image processing and analysis; Cameras (Acquisition of images); Videocon camera (Working principle & construction); Applications of Robot vision system: Inspection, Identification, Navigation & serving

Robot Programming : Teach Pendant Programming; Lead through programming; Robot programming Languages; VAL Programming; Motion Commands; Sensor Commands; End effecter commands; and Simple programs

CO4	**Explain automation & the various applications of robots.**		
M4.01	Explain the basic elements & functions of automated system	8	Understanding
M4.02	Describe application of robots in various fields	7	Understanding
	Series Test - II	1	

Contents:

Automation: Basic elements of automated system, advanced automation functions, levels of automation

Industrial Applications: Application of robots in machining; welding; assembly and material handling.

Text/ Reference:

T/R	Book Title/Author
T1	Introduction to Robotics: Analysis, Systems, Applications – Saeed B. Niku, Pearson Education
R1	Industrial Robotics: Technology, Programming and Applications – M.P. Groover, Tata McGraw Hill Co, 2001.
R2	A Text book on Industrial Robotics – Ganesh S. Hedge, Laxmi Publications Pvt. Ltd., New Delhi,
R3	2008.
R4	Elements of Robotics Process Automation, Mukherjee, Khanna Publishing House, Delhi, 2018

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/112101098/
2	https://www.robotics.org/Beginners-Guide
3	https://www.ieee-ras.org/educational-resources-outreach/educational-material-in-robotics-and-automation
4	https://www.classcentral.com/report/robotics-free-online-courses/
5	https://nptel.ac.in/courses/112105249/